

VARNA, Bulgaria

WMO: 155520

Lat: 43.232N

Lon: 27.825E

Elev: 70

StdP: 100.49

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-6.8	-12.9	1.2	-8.2	-10.1	1.6	-5.3	15.5	-2.1	13.9	-1.4	5.7	0	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	32.0	22.1	30.6	22.0	29.1	21.6	24.6	29.2	23.6	28.2	22.7	27.3	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.0	27.4	22.1	16.9	26.6	21.1	15.9	25.6	74.7	29.2	70.9	28.2	67.6	27.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.9	8.3	DB	-12.7	35.5	2.7	2.4	-14.7	37.2	-16.2	38.6	-17.8	39.9	-19.7	41.6	(4)
(5)			WB	-13.1	26.3	2.7	0.9	-15.1	26.9	-16.7	27.5	-18.2	28.0	-20.2	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	2.3	3.8	7.0	11.2	16.3	20.9	23.4	23.5	19.0	13.6	8.9	3.9	(6)	
	DBStd	8.32	4.81	4.80	4.03	3.39	3.04	2.64	2.15	2.42	3.01	3.74	4.44	4.66	(7)	
	HDD10.0	833	240	179	109	25	1	0	0	0	0	13	71	195	(8)	
	HDD18.3	2467	497	408	351	214	76	8	1	1	29	152	284	448	(9)	
	CDD10.0	1876	2	4	17	61	197	327	415	417	269	123	37	6	(10)	
	CDD18.3	467	0	0	0	1	14	84	157	159	48	4	1	0	(11)	
	CDH23.3	3840	0	0	3	13	95	625	1356	1421	298	27	2	0	(12)	
	CDH26.7	1137	0	0	0	2	11	147	437	487	51	2	0	0	(13)	
(14) Wind	WSAvg	3.4	3.6	3.8	3.7	3.4	3.1	3.1	3.2	3.3	3.4	3.4	3.3	3.7	(14)	
(15) Precipitation	PrecAvg	534	44	32	41	40	42	48	45	33	56	57	49	48	(15)	
	PrecMax	874	109	95	111	117	101	198	110	115	154	205	164	158	(16)	
	PrecMin	277	3	3	0	8	15	11	1	1	3	2	3	0	(17)	
	PrecStd	132	34	24	24	21	23	35	32	27	41	47	37	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.1	22.9	25.1	28.1	32.0	34.2	34.8	30.2	26.2	22.9	17.2	(19)	
		MCWB	11.1	12.4	12.7	16.0	19.6	22.2	21.9	22.4	19.9	19.1	15.9	12.6	(20)	
	2%	DB	13.1	15.2	18.8	21.8	25.8	29.9	31.9	32.2	28.1	23.3	19.8	15.0	(21)	
		MCWB	9.3	10.4	11.8	14.4	18.2	21.5	22.2	22.1	20.1	17.6	15.1	11.4	(22)	
	5%	DB	10.8	12.8	16.0	19.1	24.1	28.2	30.2	30.9	26.5	21.3	17.2	12.8	(23)	
		MCWB	8.1	9.2	10.4	13.3	17.7	21.0	22.1	22.2	19.9	16.8	14.1	10.3	(24)	
	10%	DB	8.8	10.2	13.2	17.0	22.4	26.9	29.1	29.2	25.0	19.8	15.2	10.9	(25)	
		MCWB	7.3	7.5	8.9	12.3	17.0	20.6	21.7	21.9	19.3	16.0	12.9	9.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.9	13.9	16.8	21.1	24.5	25.6	26.3	23.4	20.6	16.9	13.7	(27)	
		MCDB	14.9	17.1	20.6	23.1	25.4	29.3	30.2	30.3	27.0	24.3	20.6	16.6	(28)	
	2%	WB	9.9	10.9	12.2	15.1	19.8	23.1	24.4	24.9	22.0	18.8	15.7	12.1	(29)	
		MCDB	12.5	14.5	17.9	20.4	24.1	27.6	29.1	29.5	25.7	21.9	18.7	14.3	(30)	
	5%	WB	8.5	9.3	10.7	13.9	18.7	22.2	23.5	23.8	21.0	17.6	14.6	10.6	(31)	
		MCDB	10.2	12.2	15.0	18.2	22.7	26.6	28.3	28.3	24.5	20.4	17.1	12.1	(32)	
	10%	WB	7.1	7.8	9.6	12.8	17.7	21.2	22.6	22.9	20.1	16.4	13.2	9.2	(33)	
		MCDB	8.4	9.7	12.9	16.3	21.4	25.6	27.3	27.4	23.7	18.9	15.0	10.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	9.8	10.3	10.7	10.9	11.4	11.8	10.8	9.7	8.4	8.1	(35)	
		MCDBR	11.4	13.2	14.3	13.7	13.1	12.8	12.9	13.3	12.8	12.6	11.6	10.8	(36)	
	5% WB	MCWBR	7.8	8.4	8.1	7.5	7.4	7.0	6.9	7.1	7.1	7.7	7.2	7.5	(37)	
		MCWBR	10.0	12.3	13.0	12.5	11.9	11.7	11.9	12.2	11.8	11.7	10.3	9.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.351	0.381	0.415	0.460	0.433	0.436	0.445	0.455	0.432	0.398	0.373	0.353	(40)	
		taud	2.327	2.247	2.179	2.064	2.225	2.254	2.243	2.210	2.241	2.312	2.322	2.329	(41)	
	Ebn at Noon		761	798	816	807	843	839	827	804	791	770	726	718	(42)	
		Edh at Noon	85	108	130	158	138	135	135	135	121	100	85	78	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.43	2.16	3.35	4.78	6.02	6.61	6.66	5.97	4.29	2.70	1.62	1.25	(44)	
	RadStd		0.19	0.27	0.33	0.50	0.39	0.46	0.39	0.30	0.36	0.27	0.22	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-102	+94	+50	(46)	
(47)	Regional (28 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-114	+86	+52	(47)	

Nomenclature: See separate page